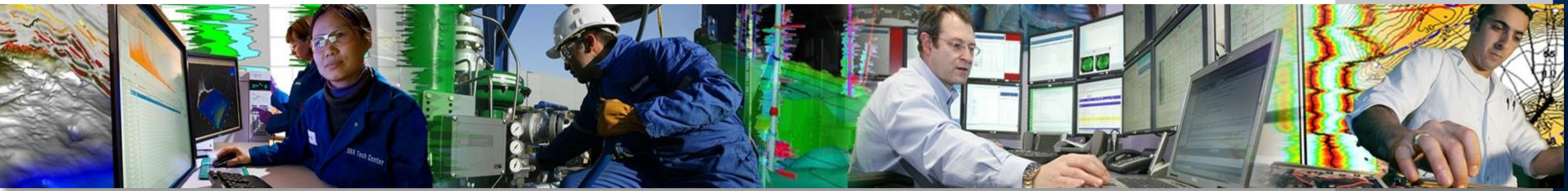
Mapping the Subsurface
- with Petrel



Well Correlation in Petrel – Setup and Process

A Schlumberger Company

NExT

Network of Excellence in Training

Objectives

- Create Log Property Templates
- Create and Display Well Sections
- Maneuver – Scroll and Zoom
- Curve Filling – Group Panels, Cut Offs, etc
- Well Templates
- Well Tops – Create, Edit, Flatten, etc.
- Ghost Curves
- Log Calculator
- Interactive Facies Interpretation
- Extra:
  - Completion Design, Comment Logs, Well Seismic and Bitmap Logs

NExT

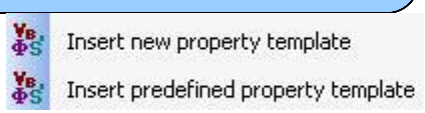
Create or Edit Log Templates

The standard Petrel templates may not always fit your needs, so they can be changed or you can make new ones.

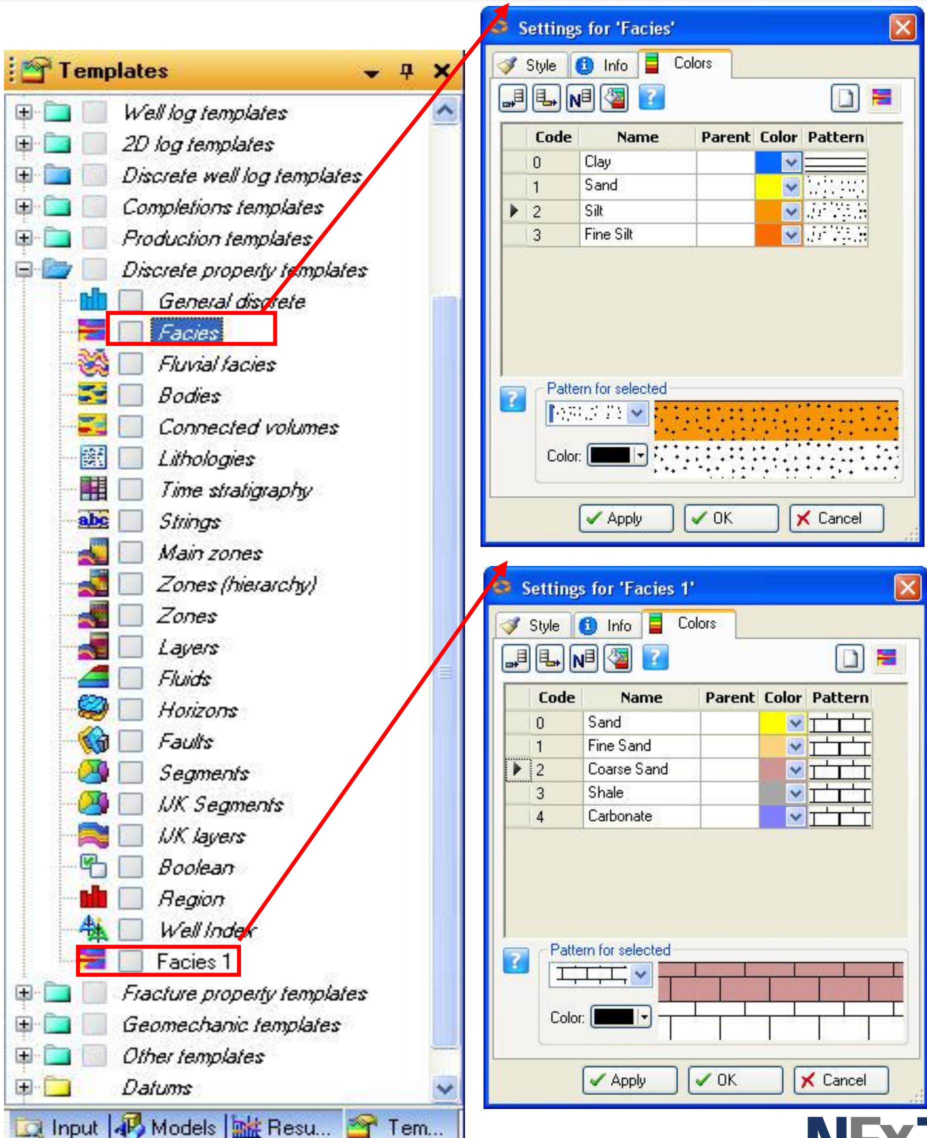
Go to the **Templates pane**.

Copy an existing template using Ctrl+C/Ctrl+V, and edit it in the **Colors tab**.

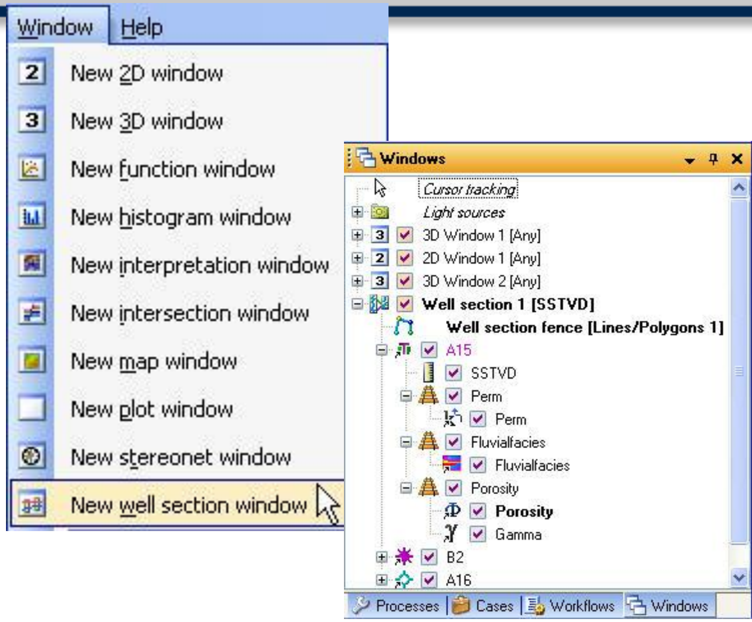
To create a new template, right-click on the folder where it should reside and select a new template or a predefined one:

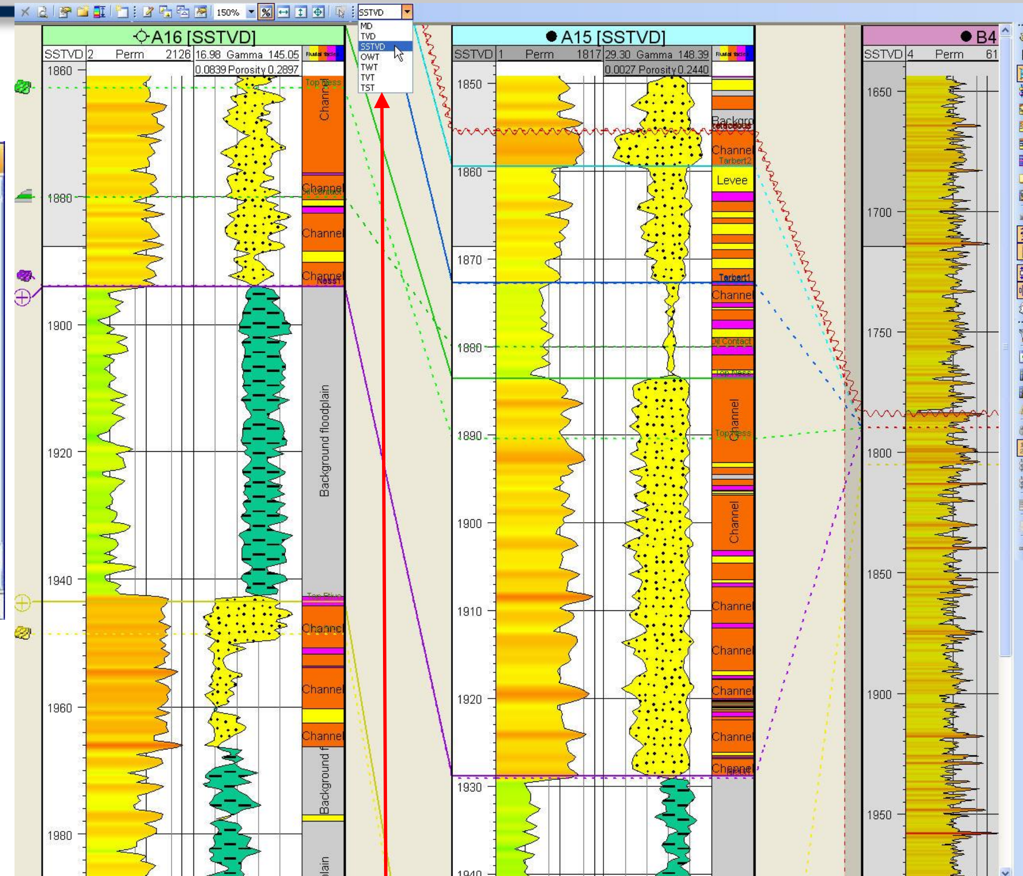


Note: These are global templates, any change will affect all objects using the same template.



Open Well Section and View Data





Well Sections are generated from the **Windows menu**...

...and stored in the **Windows pane**

Click on the check box in front of a Well Section to display existing Well Sections

To choose a depth scale, select between MD, TVD, SSTVD, OWT, TWT, TVT and TST

Display Well Tops, Fault Cuts, Surfaces, Horizons and Fluid Contacts

NExT

Creating Well Section and Displaying Data

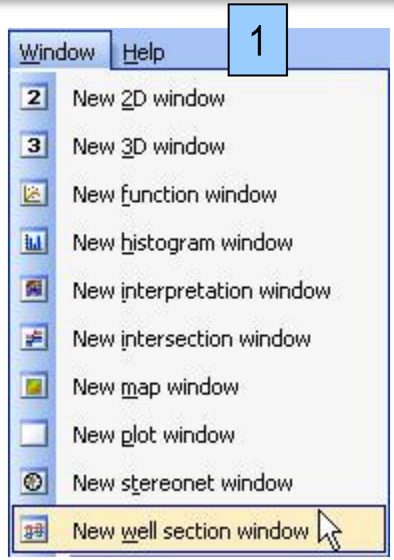
1. Insert a new Well Section from the **Windows Menu** (previous slide). An empty well section window opens.

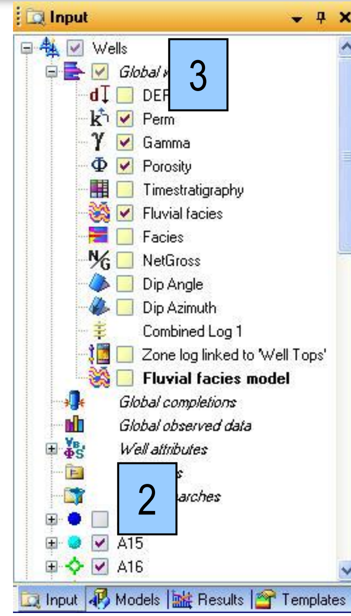
2. Go to **Input pane** and select the desired Wells.

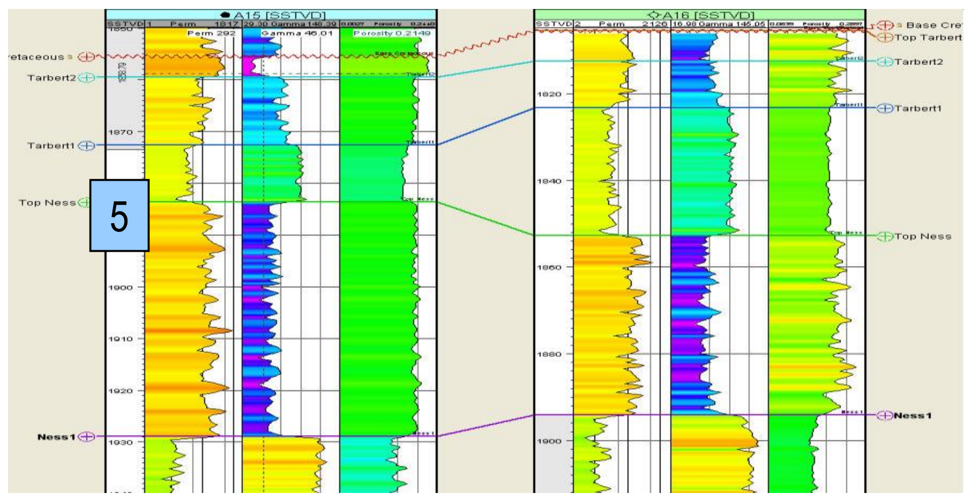
3. Go to **Global Well Logs folder** and display the logs of interest.

4. Go to **Well Tops folder** and click on the Well tops .

5. Selected objects are displayed using default parameters in the **Well Section Window**. Change these to fit the a preferred set up.





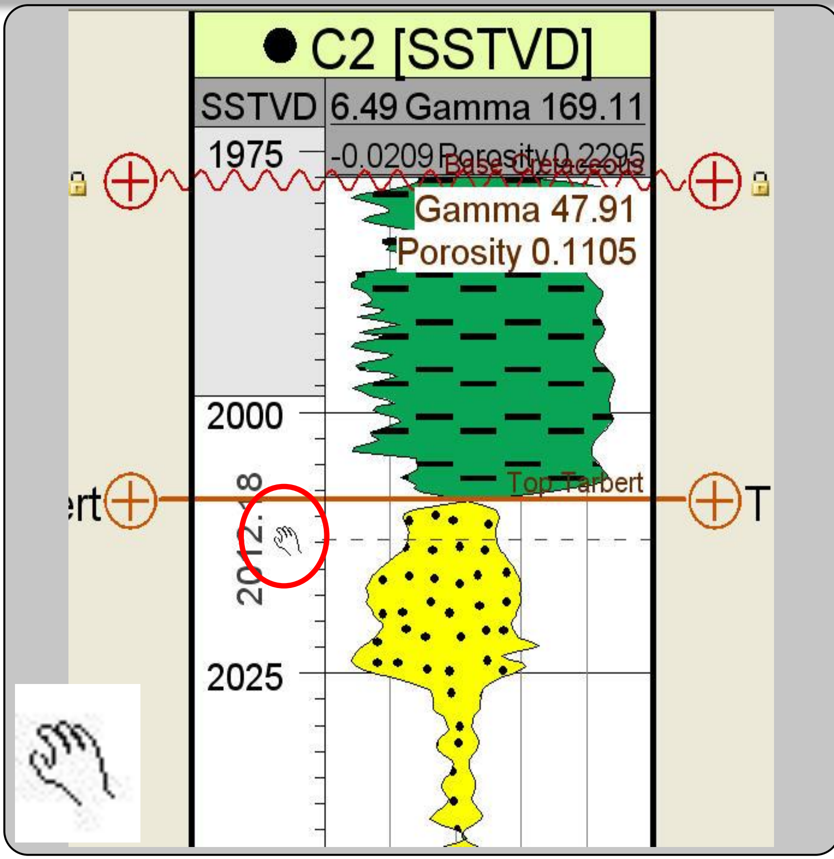


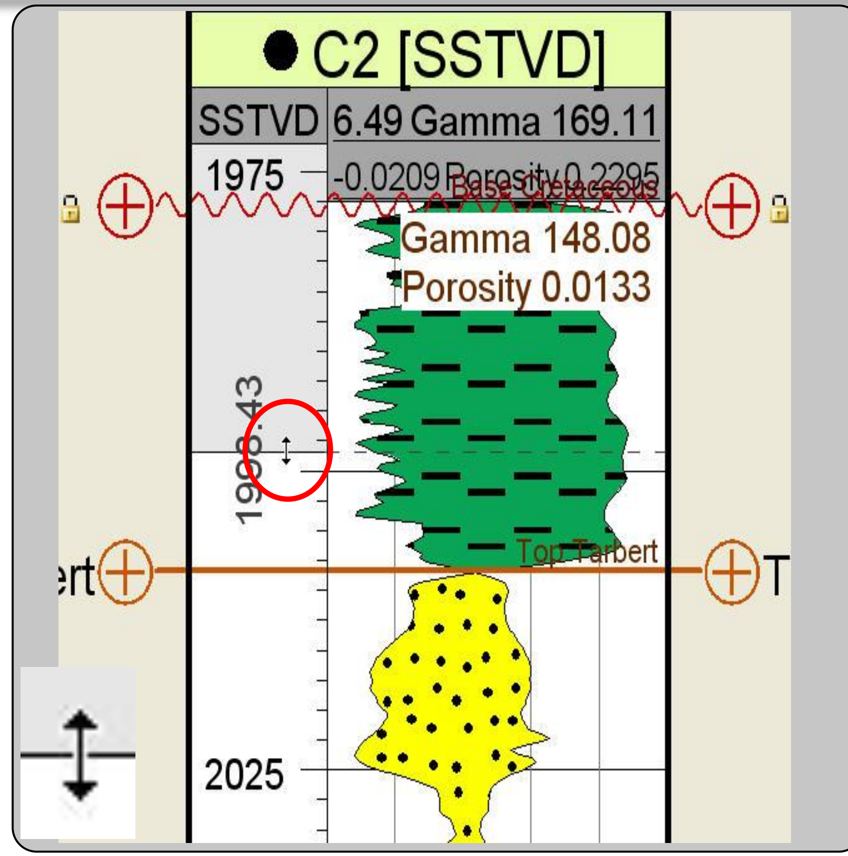
NEXT

Scrolling and Scaling – Interactive

Synchronize scrolling

Synchronize scaling





Position cursor on the white depth panel. Move the hand icon while using left mouse button to scroll display up or down the well.

Position cursor on line between gray and white area. Move the arrow icon to stretch or squeeze the well.

NExT

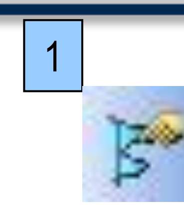
Curve filling – Single Curve Panel

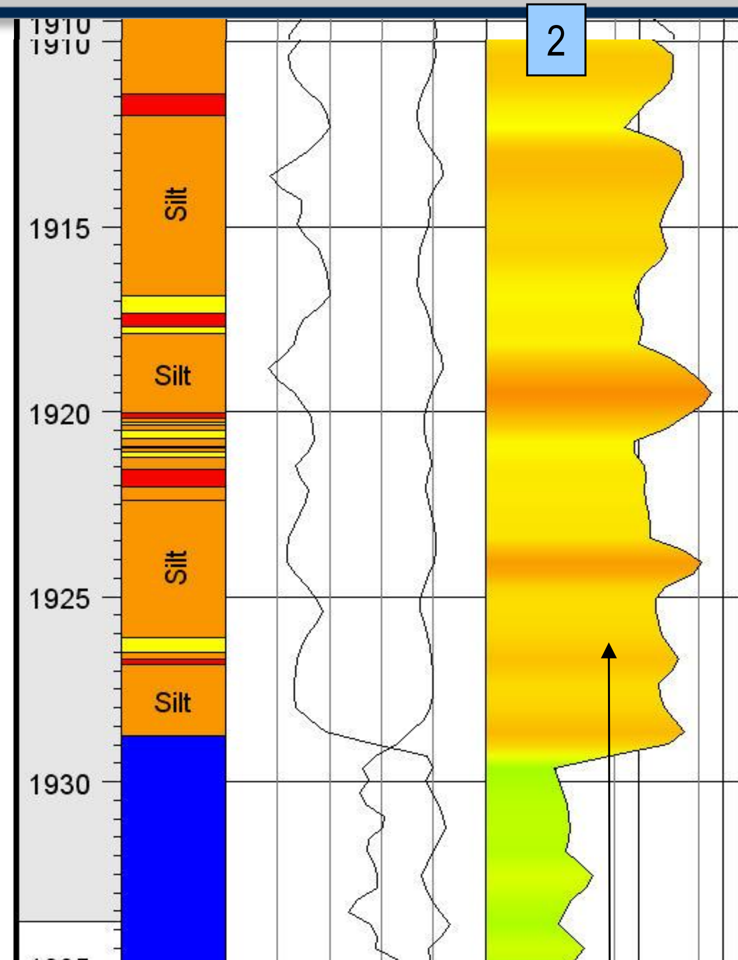
How to color fill a Single curve panel:

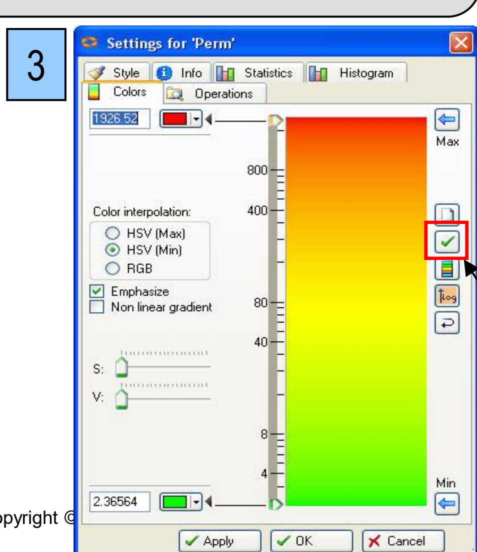
1. Click the **Create/Edit Curve Fill** icon from **Function bar**.

2. Click with the left mouse button in area between log curve and panel edge.

3. To adjust the color range, open Settings for the log from **Global Well Logs**, go to the **Colors tab.**







Note: The Permeability color fill template can have a logarithmic scale

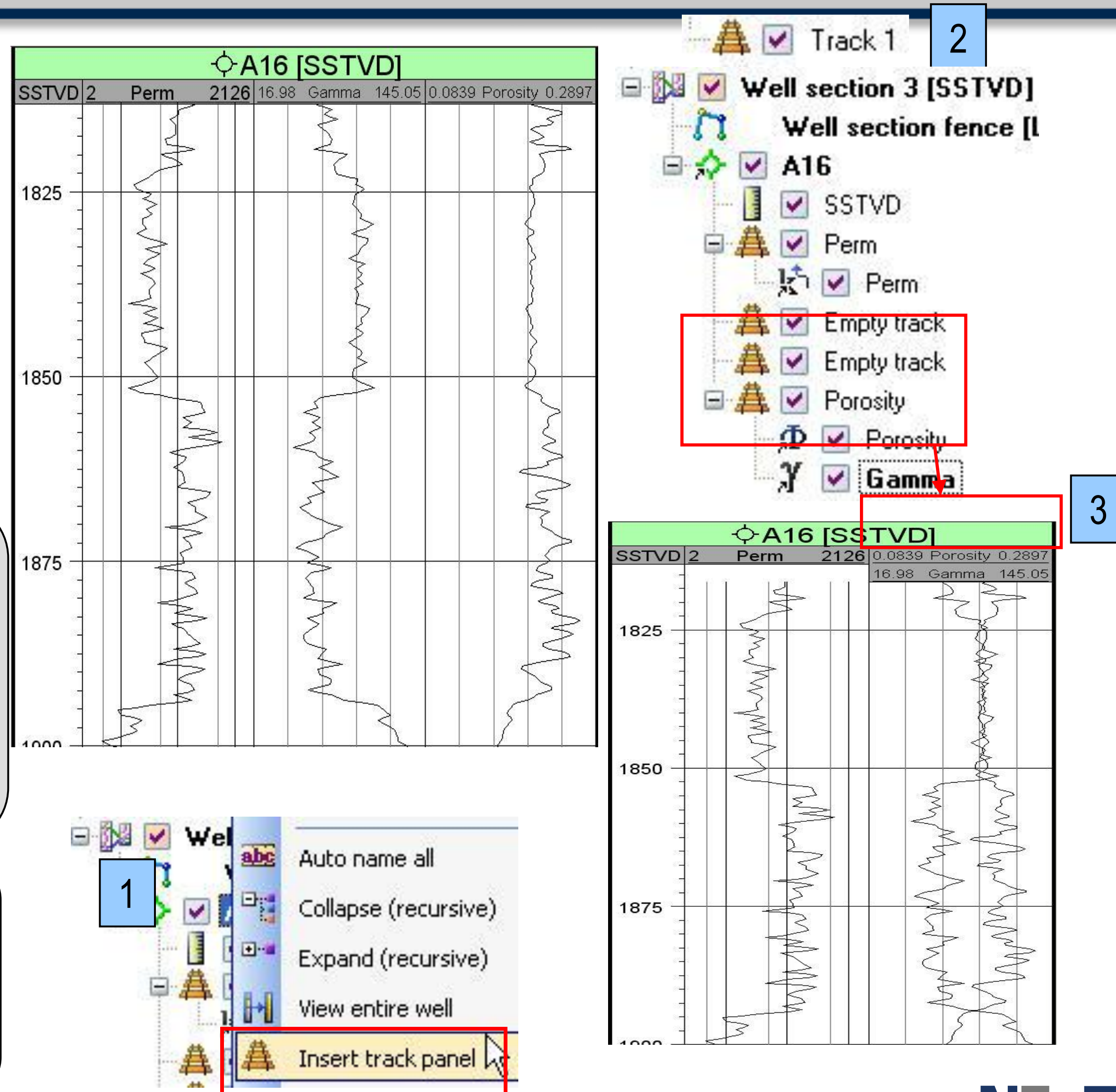
Setting up Track Panels

Track panels allow you to display one or more logs in the same panel.
Preparation:
Make sure logs are posted in Well Section, without color fill.

1. Right-click on a well in the Well section in the **Windows pane** and select **Insert track panel**.

2. A folder called **Track 1** is created. Drag-and-drop curves into the track folder (left mouse button). Name of the track automatically updates according to the log dropped into the track.

3. In the Well Section Window there are now multiple logs in one panel. Old log tracks are now called Empty track.



NExT

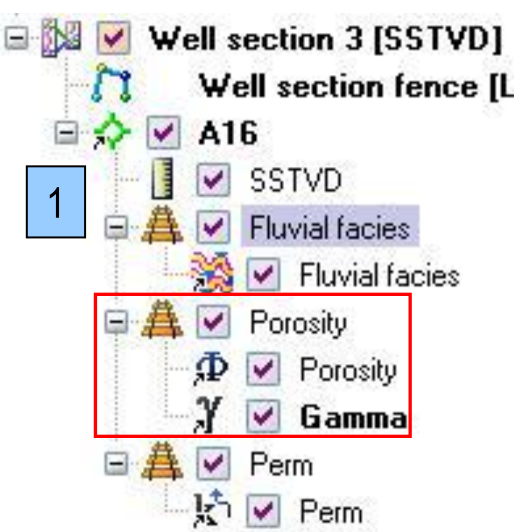
Fill Between Two Curves – In a Track Panel

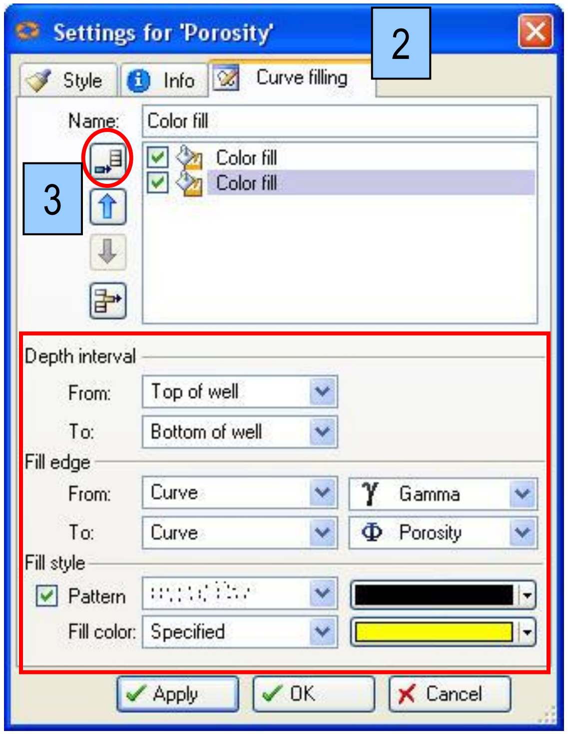
1. Expand the Track folder. Double-click on the track name, where the two logs are located, to open the Settings dialog.

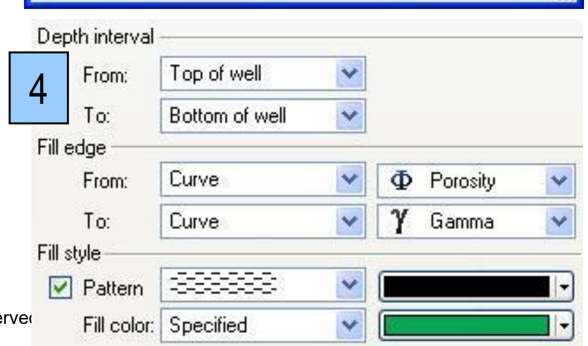
2. Go to **Curve Filling tab**.

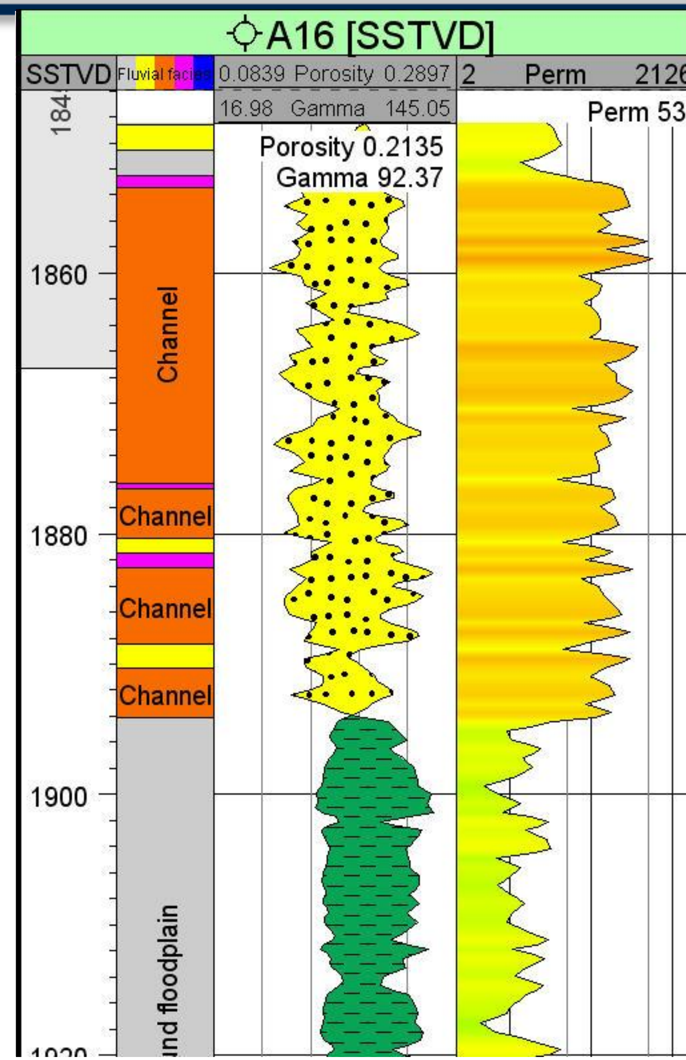
3. Append two new Color fills, define depth interval, fill edge and fill style.

4. Make specification for both curve fills. **Apply.**







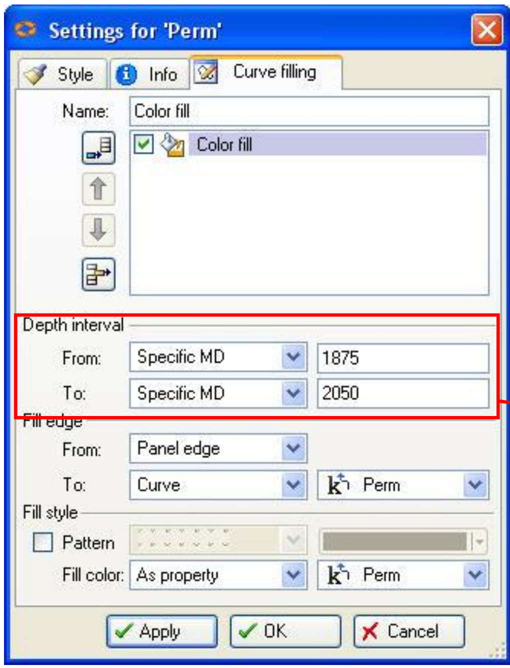


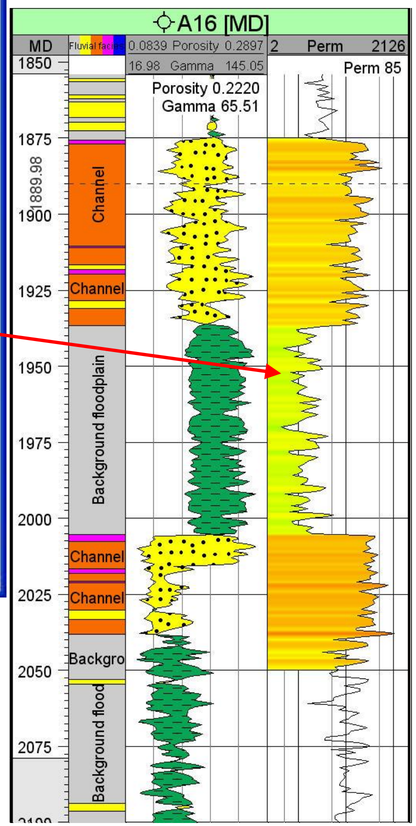
NExT

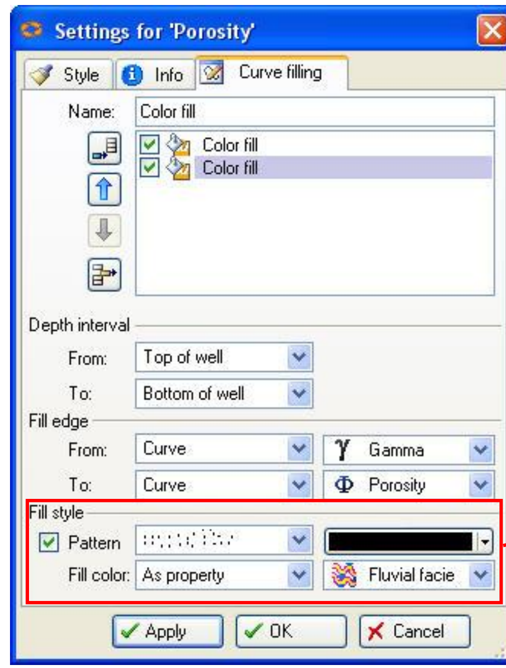
Curve Filling – Cut Offs and Property Fill

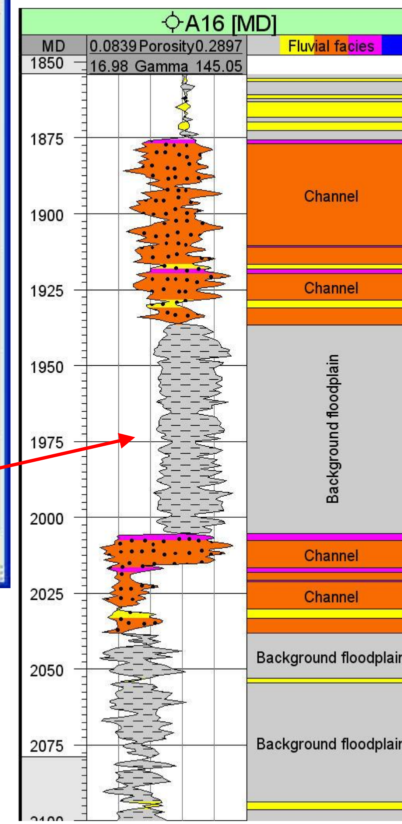
Setting a Cut Off:
Specify MD Depth Interval and Color fill from a base line to a curve or a constant.

Automatic Property Fill:
Color fill using a property to define the color range. Select As property and specify the property.





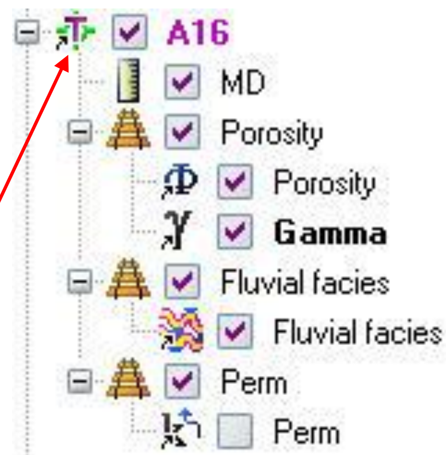


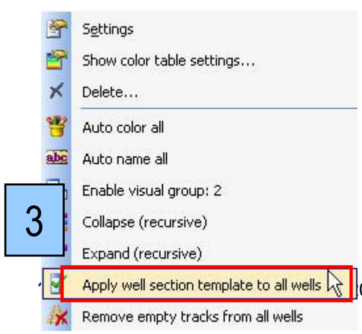


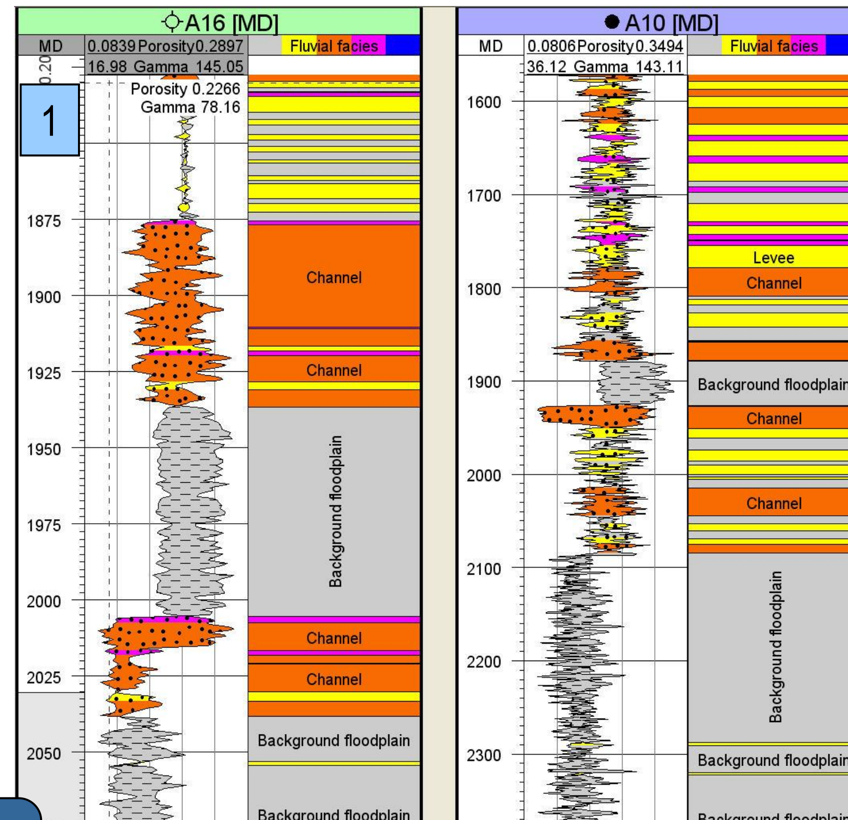
Creating and Applying Well Templates

1. Define a template: set up a Well (logs, fill, groups etc.).
2. Right-click on the Well and select **Use as well template** (Well turn purple).

3. To apply template to all wells in section; right-click on the Well Section and select **Apply well template to all wells.**
4. Choose the template from the drop-down menu and click **OK**. Wells in section are updated.





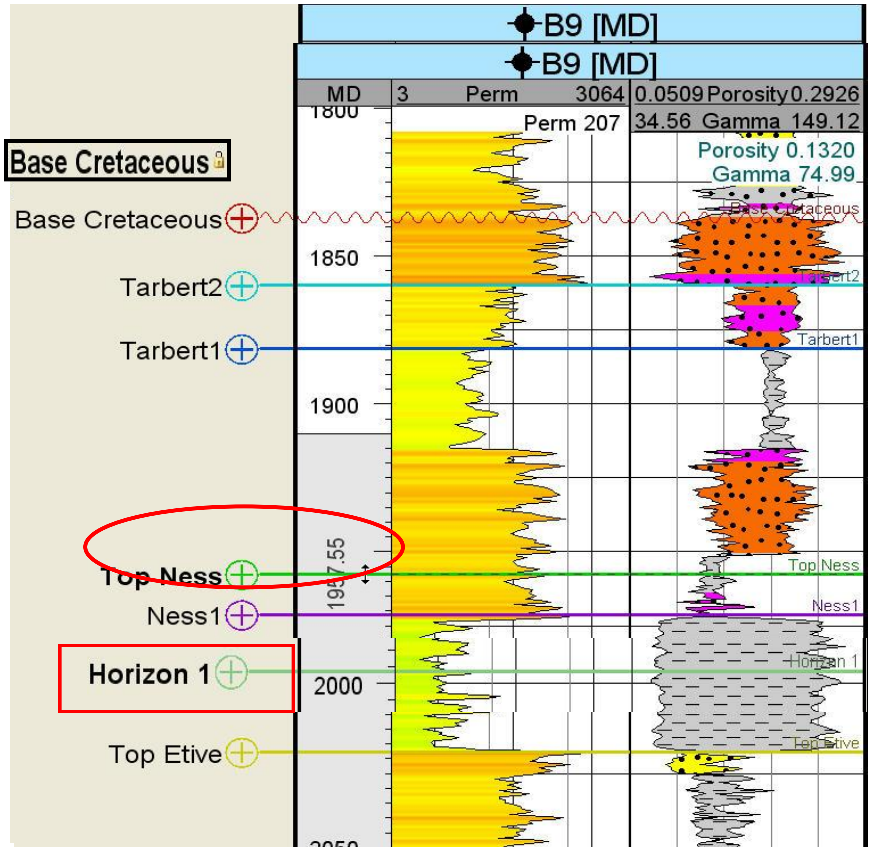


Use **Apply well template** to select a template for a single well.

NExT

Well Tops – Create and Edit

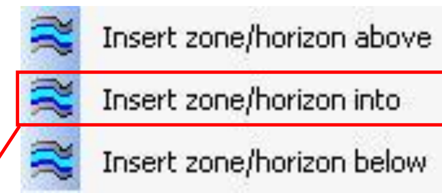
Edit well tops



Create well tops

...

or



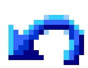
All changes are reflected in the well tops ***spreadsheet***

Well top spreadsheet for 'Well Tops'

Well filter B9

	Well	Surface	X	Y	Z	MD
3	B9	Base Cretaceous	456719.08	6785550.13	-1836.19	1836.97
86	B9	Tarbert2	456717.47	6785548.40	-1858.92	1859.81
72	B9	Tarbert1	456715.91	6785546.78	-1880.27	1881.29
27	B9	Top Ness	456713.52	6785544.23	-1914.37	1915.56
57	B9	Ness1	456709.98	6785540.39	-1969.82	1971.26
88	B9	Horizon 1	456708.45	6785538.52	-1994.80	1996.35
40	B9	Top Etive	456707.21	6785536.96	-2016.09	2017.73

Undo well top edit (once)

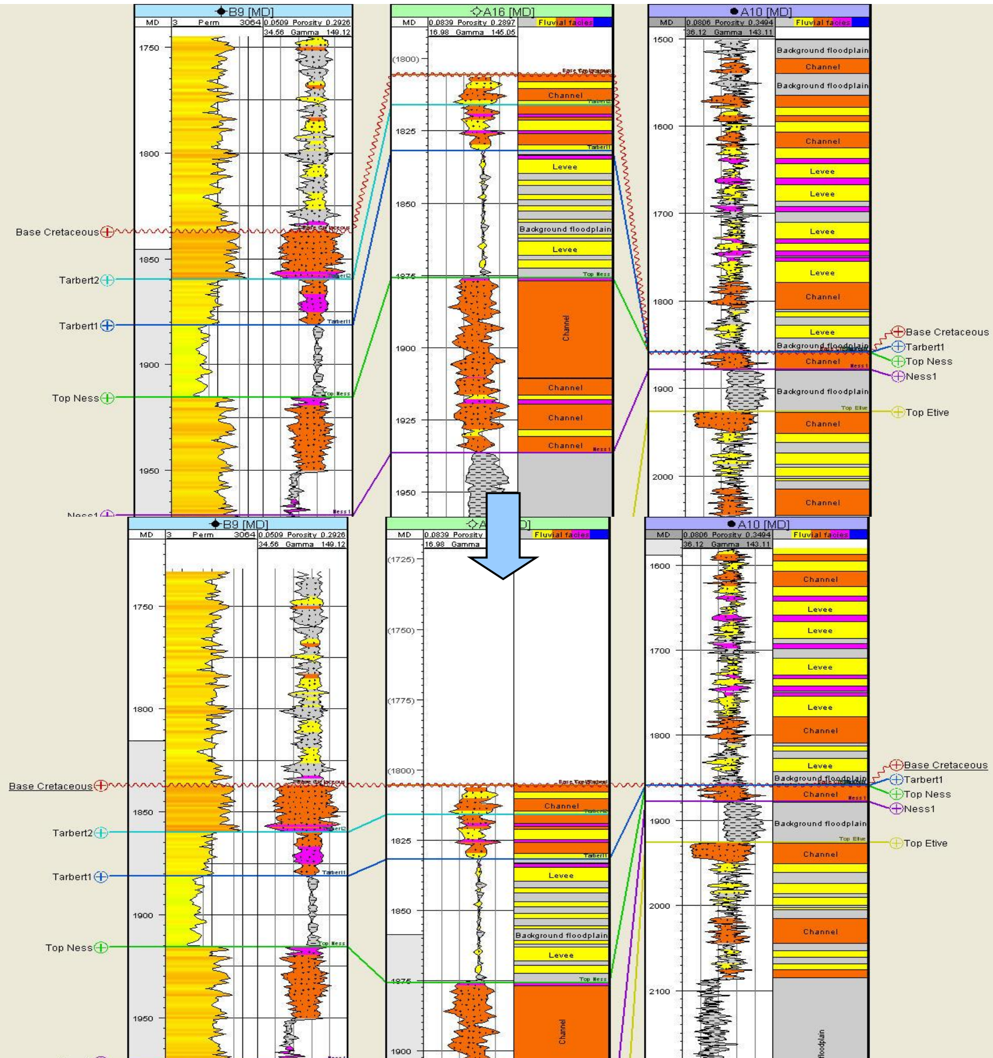


Lock a well top from its Settings dialog to avoid mis-edits

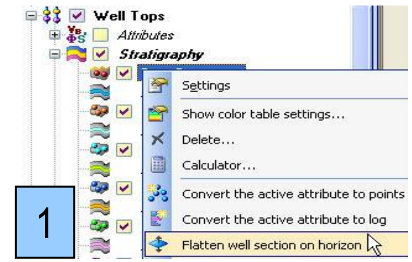
☑ Lock well top

Well Tops – Flattening

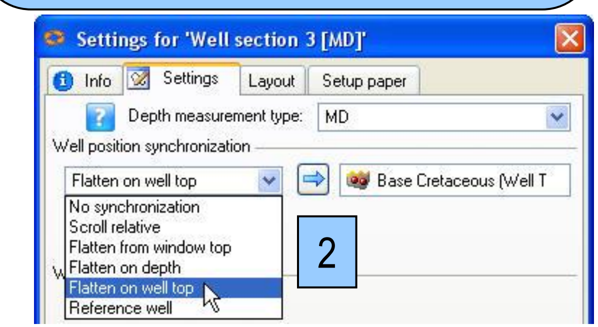
Flattening on Well Tops is used to correlate intervals. It can be done in 2 ways:



1. Right-click on a Well top in **Well Tops folder** and Select **Flatten well section on horizon.**



2. Go to **Settings tab** in the settings for **Well Section**; choose **Flatten on well top,** highlight a well top, drop it into the field using the blue arrow. **OK**.

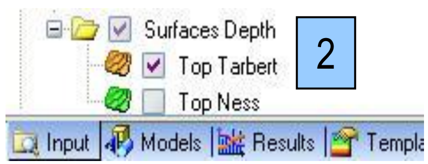


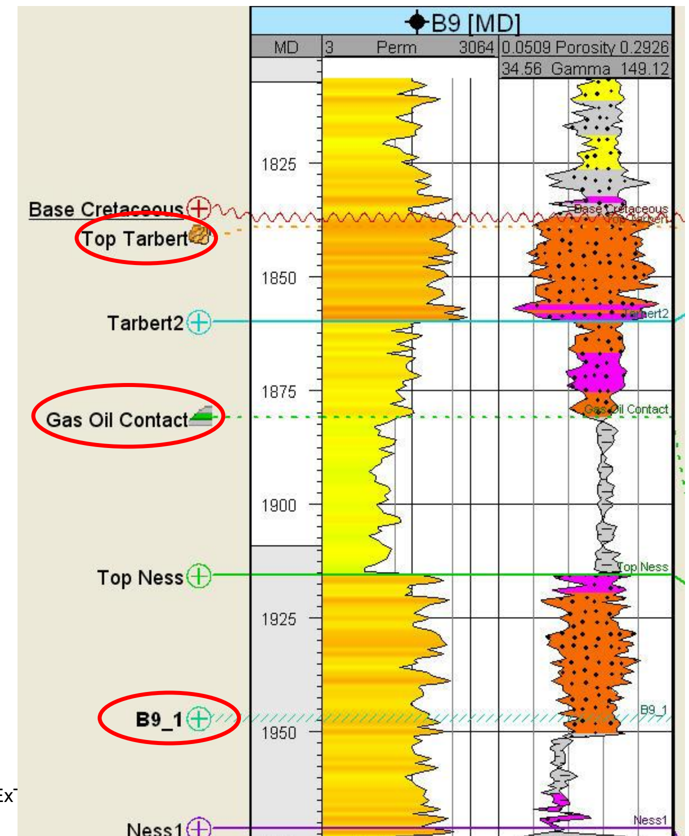
Contacts, Surfaces and Fault Gaps

1. Fluid Contacts

2. Surfaces and Horizons

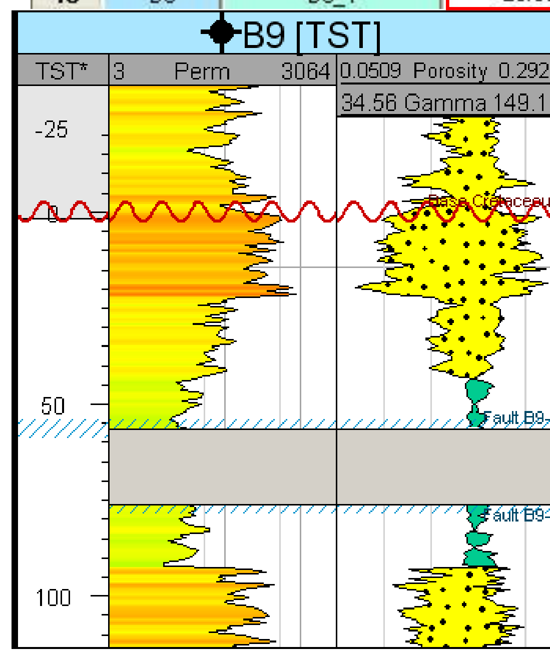
3. Faults (MD, TVD)





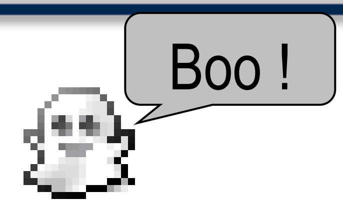
Fault gaps (TST, TVT); define throw in the Well Tops Spreadsheet under **Missing**

	Well	Surface	Missing
33	B9	Base Cretaceous	0.00
34	B9	Tarbert2	
35	B9	Tarbert1	
36	B9	Top Ness	
37	B9	Ness1	
38	B9	Horizon 1	
39	B9	Top Etive	
40	B9	B9_1	-20.00



Ghost Curve – Create

Ghost curve: transparent copy of a well in the well section. Can be moved independently in the well section window. Compare with the logs from other wells.

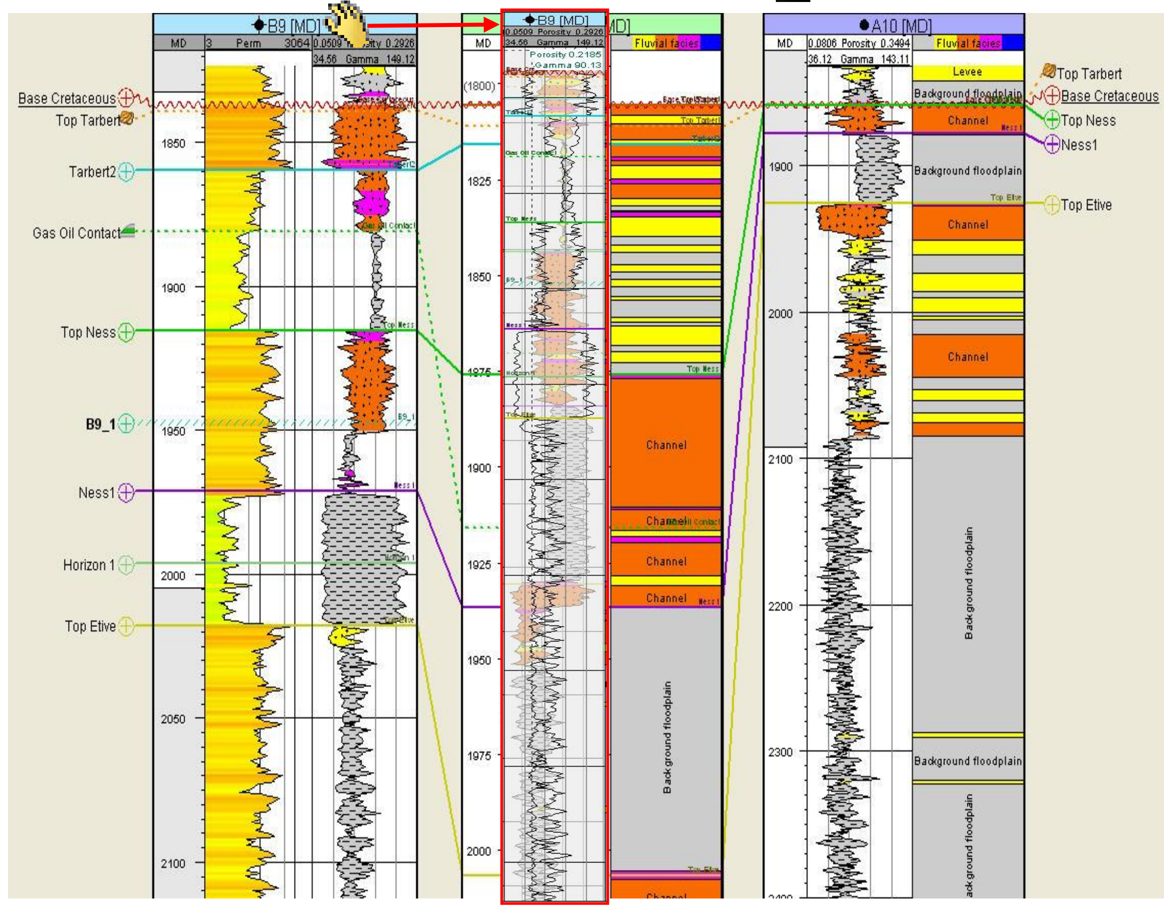


Ghost curves can be made for one curve...

Drag the desired panel from the reference well by holding the hand tool over the well name at the top. Drag using the left mouse button. Ghost icon is added to the Well section.

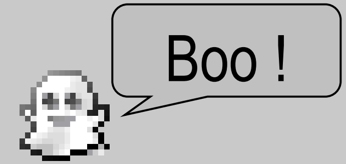
... or the entire well curve panel:

Click the Ghost icon in the **Function bar**. Position cursor on well name and drag to activate ghost. Open Settings dialog for the ghost. On the Style tab, select ***None*** for the Dynamic adaption.

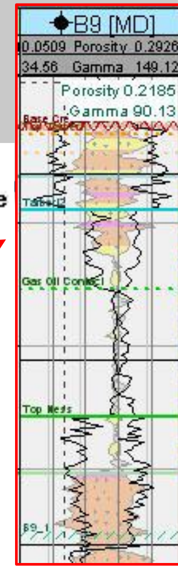


NEXT

Ghost Curve – Manipulate



1. Ghost curves are stored separately in the Well Section under the **Windows pane**.



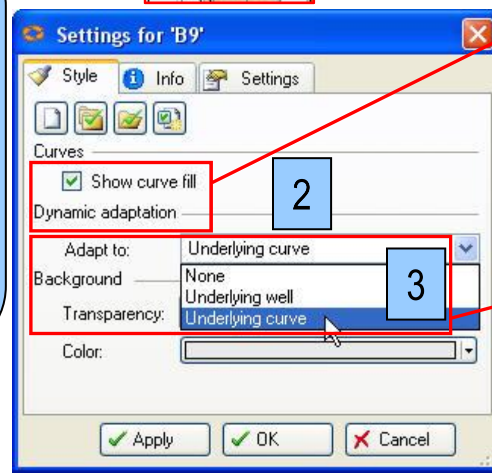
2. Settings: Curve fill and Transparency is available.

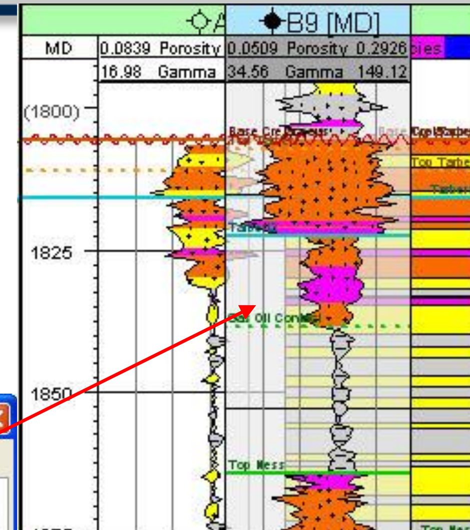
3. Dynamic adaptation under the Style tab in the settings for the Ghost curve has 3 options:

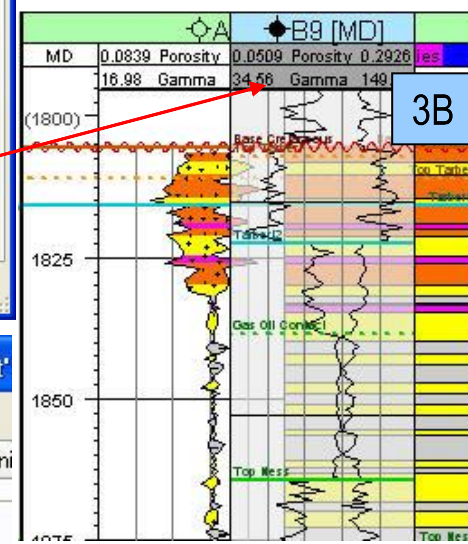
A) **None** – curve remains unchanged

B) **Underlying curve** – adapts to curves it is dragged over

C) **Underlying well** – the ghost curve changes to match the template of the underlying well







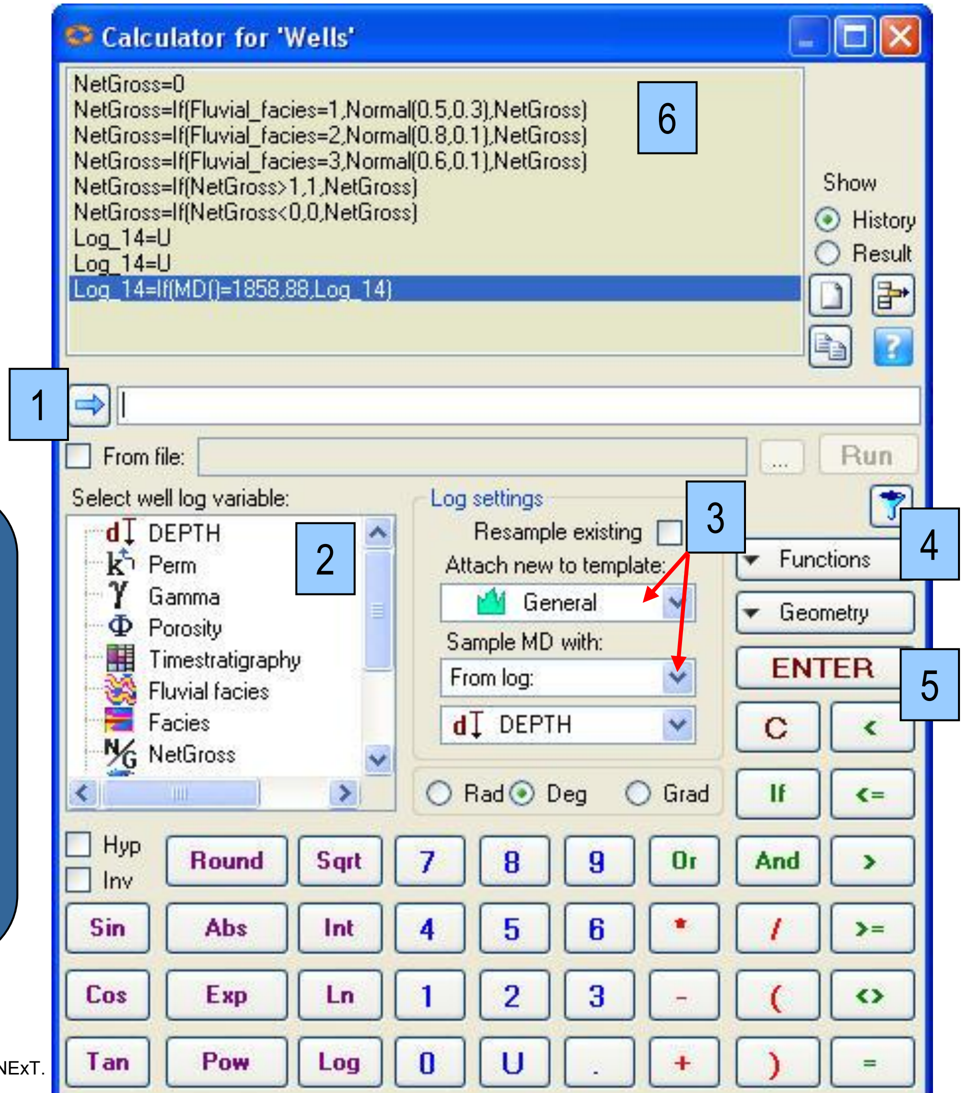
Note: If a lot of scrolling is done for correlation it is a good idea to turn off decimation of underlying curve; **Project menu>Project settings>Well settings tab**.

NExT

Log Calculator

Use the Log calculator to perform operations on or between well logs. Available wells and logs will depend on which well or log you start the calculator from.

Note: The calculator can be run on Wells, Well Subset or all Wells:
Right-click on their respective main folder.



1. Input equation.
2. List of available logs.
3. Select template for new log.
4. Functions and Geometry pull-down menus.
5. Click **Enter** key to run equation.
6. List of previously run equations.

NExT

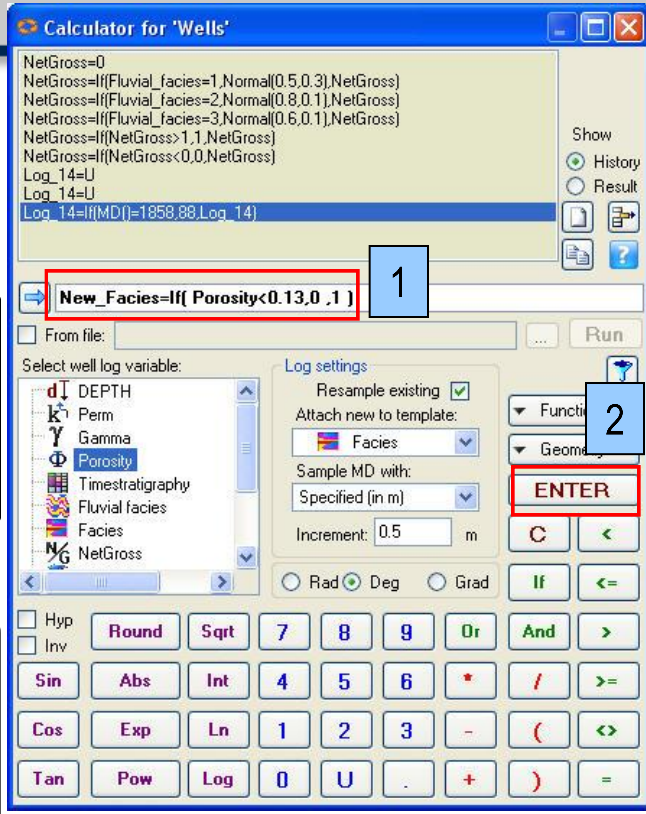
Log Calculator – Example of use

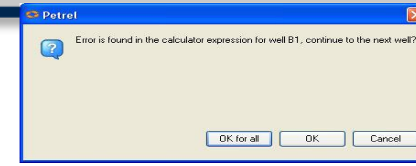
Create a new discrete log that is defined by cut-off values from a continous logs.

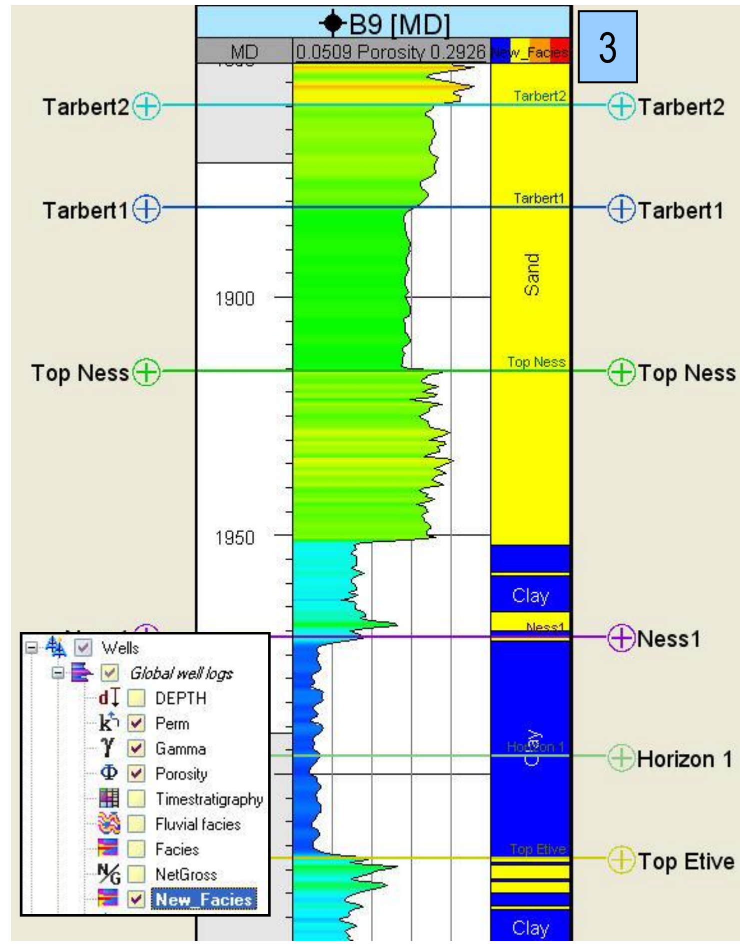
1. Statement: If porosity log is less than 13% then Facies code 0 (Clay) else if it is higher, then Facies code 1 (Sand).

2. Press **ENTER**, first error message for no log in B1, press **OK,** and second error message for expression for B1, press **OK for all**.

3. The new log is stored in the Global well logs folder. Compare in Well Section window.







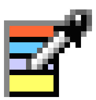
NEXT

Interactive Facies Interpretation

Edit discrete logs

Paint discrete log class

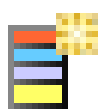
Flood fill discrete log class



Pick up discrete log class value

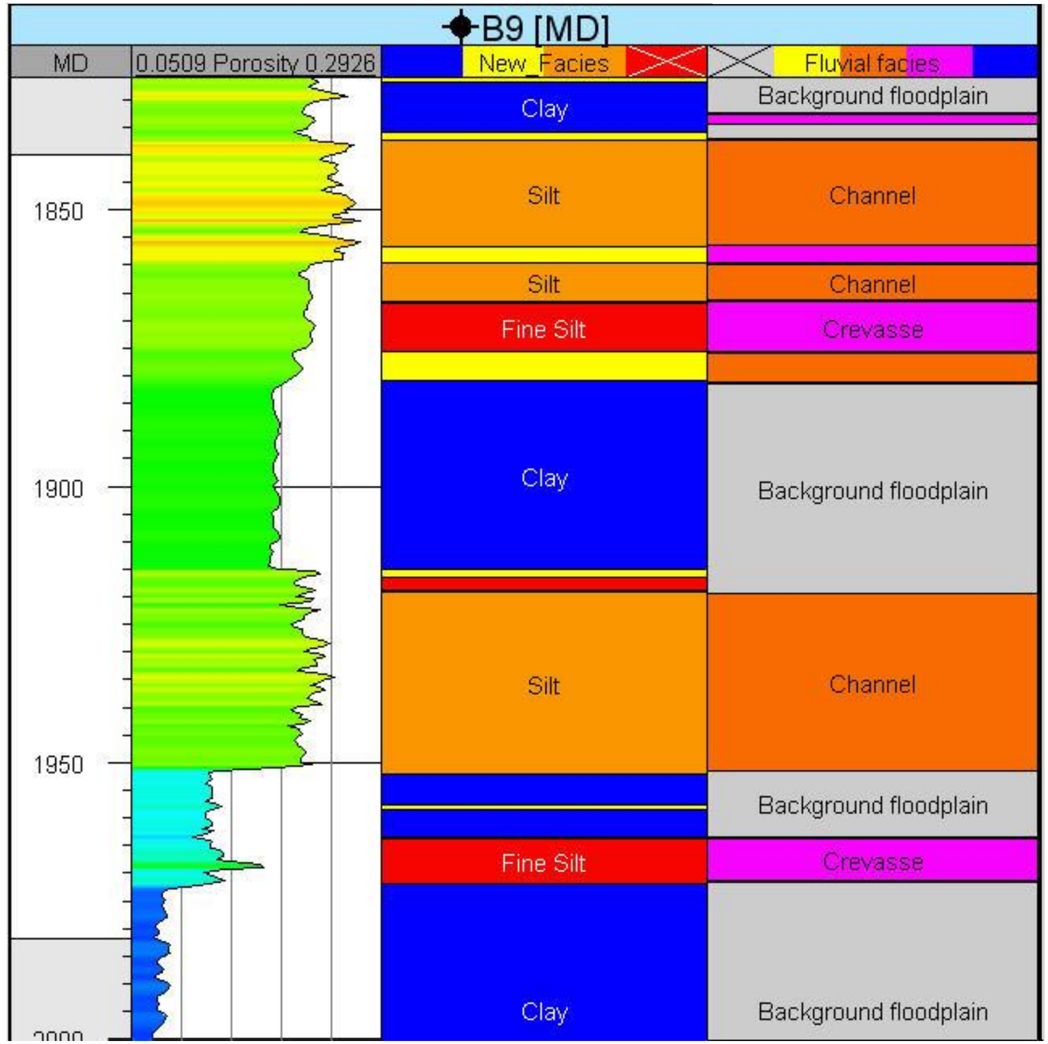
Create discrete logs to be painted

+



Paint discrete log class

Create new discrete log

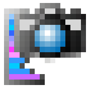


NExT

Bitmap Logs

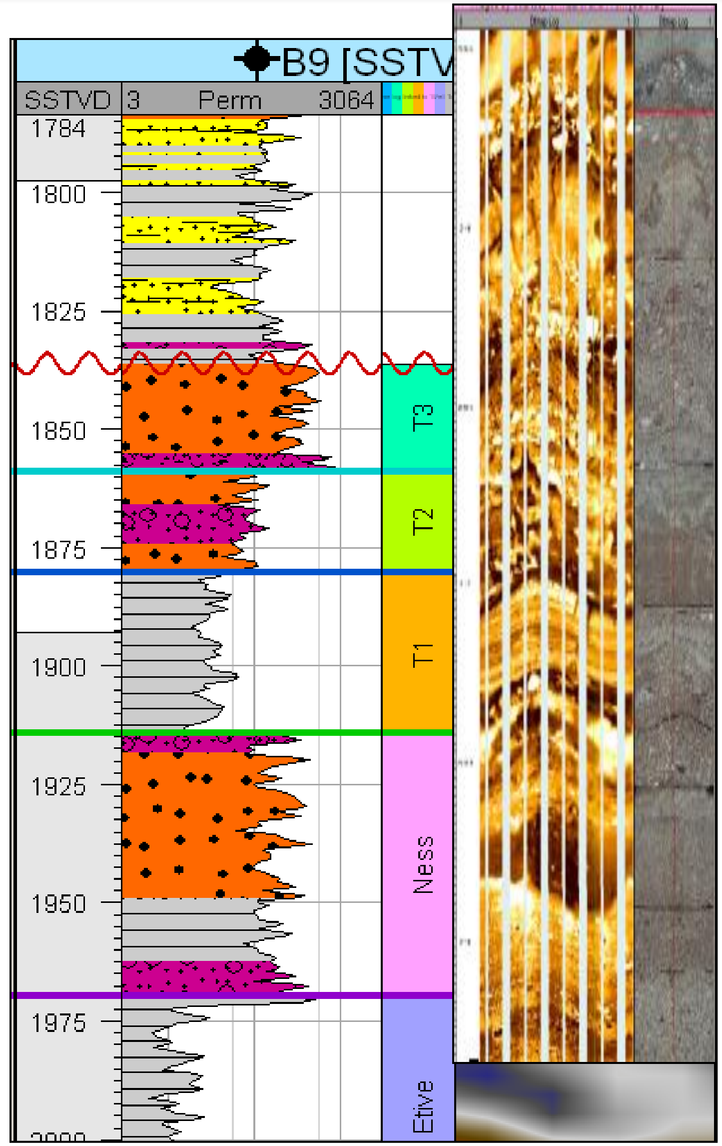
Import **Bitmap logs** such as FMI and Core images

Edit Bitmap logs:



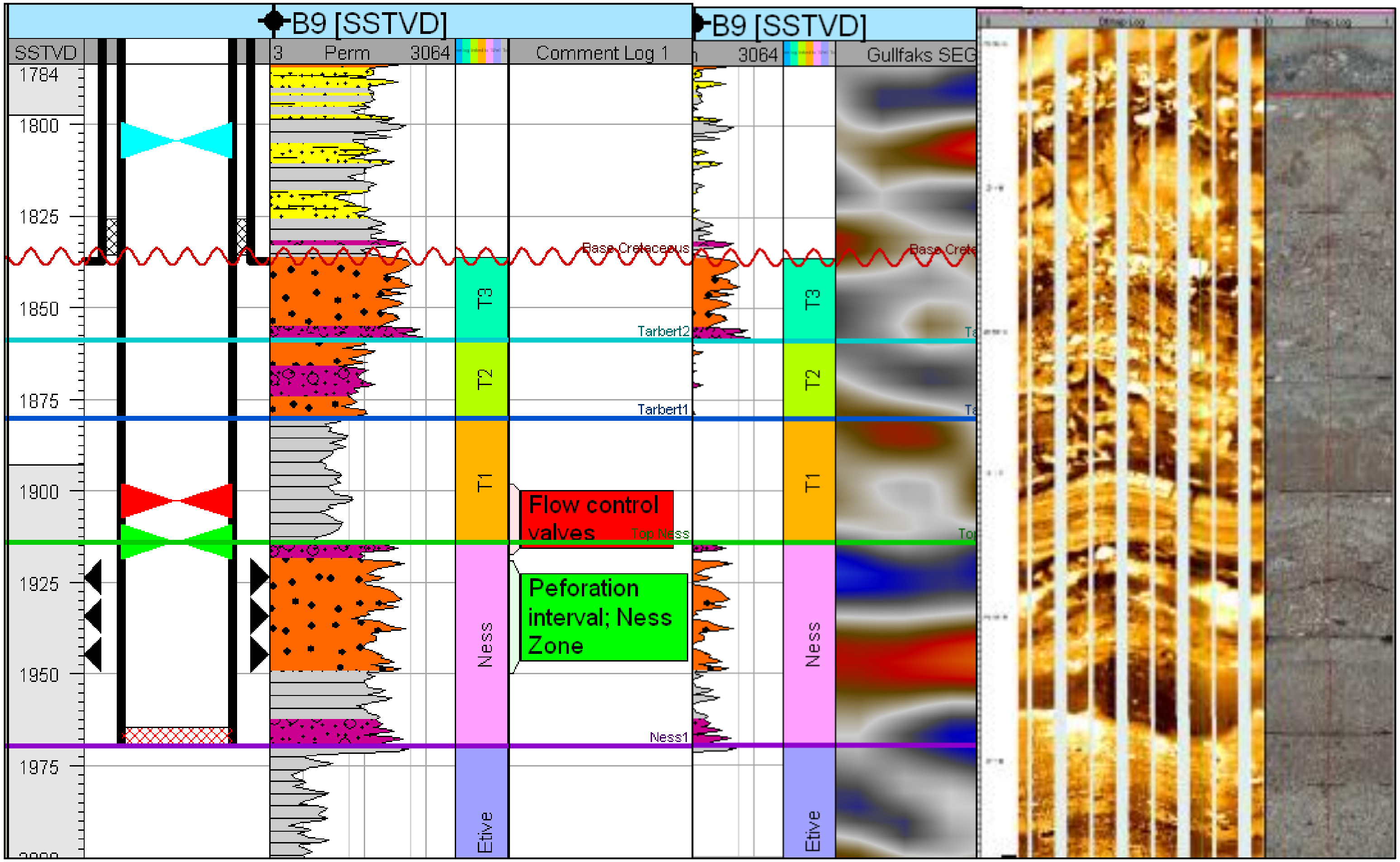
Create/edit control points

Import **Raster logs**:
Bitmap logs (A2D) with Smart Interchange Format (SIF) – relate bitmaps to depth



NExT

Completion Design, Comment Logs, Well Seismic and Bitmap Logs

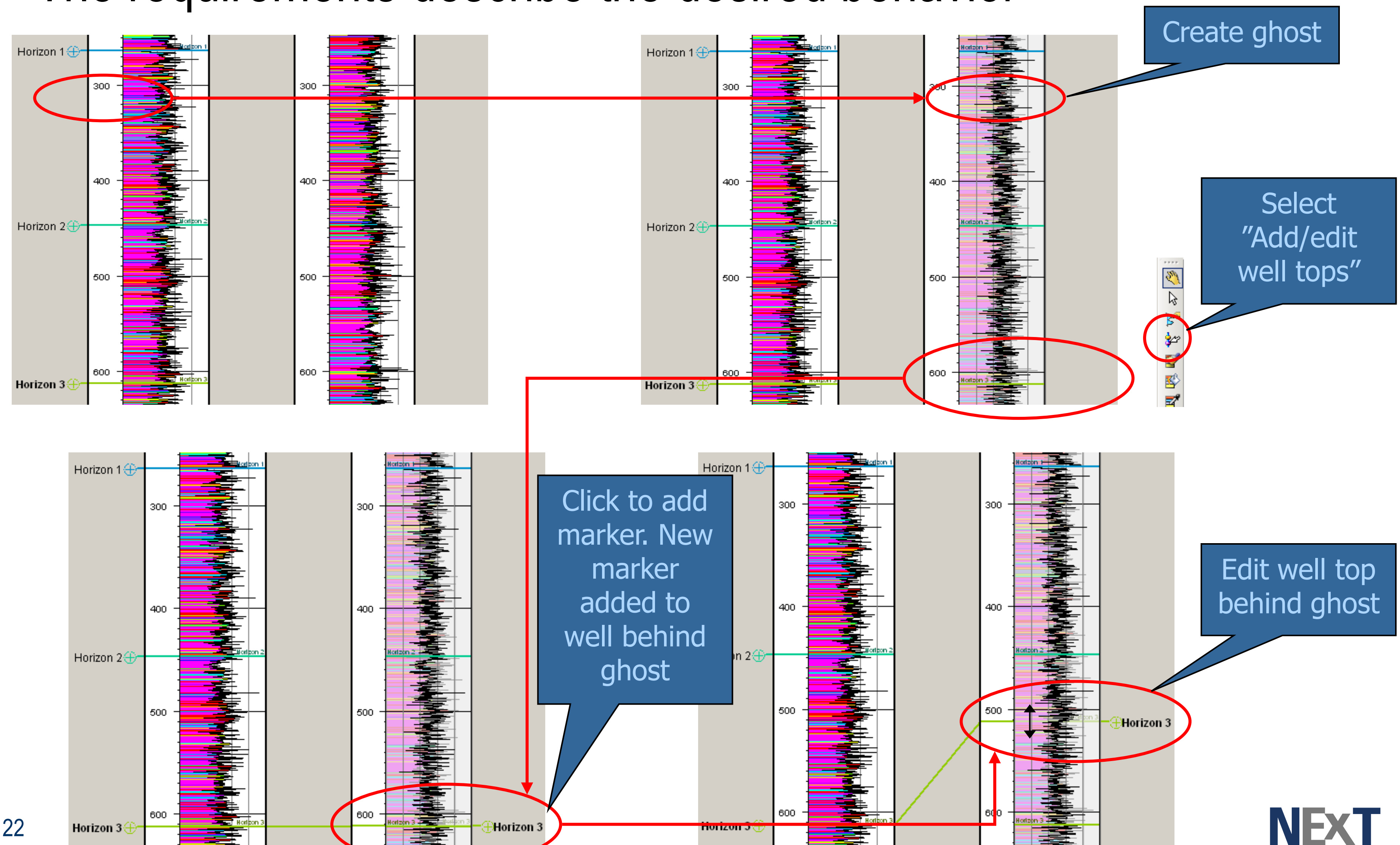


NExT

Specification

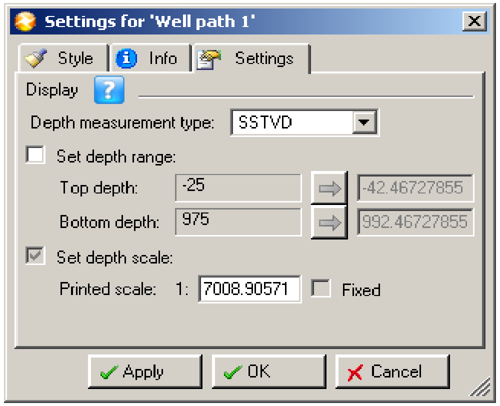
1. Add/edit markers in well behind ghost

- The requirements describe the desired behavior



Specification
Usability issues with ghost size and scale interactions

- Initialise ghost at same scale and measurement type as parent well
- Ghost scale must never change in response to changes in ghost panel size
- Well section scale must not be affected by the following:
  - Interactive rescaling of the ghost in relative or absolute ”synchronise well scaling” modes
  - User dialogue edits to printed ghost curve scale
- Ghost scale settings should not be affected by the following:
  - Interactive rescaling of wells in the well section in relative or absolute ”synchronise well scaling” modes
  - User dialogue edits to absolute well section scale
- Settings tab of ghost settings should not adapt to the underlying well
- Ghost scale should always be independent of well section. To communicate this to the user, ”Set depth scale” should be greyed out in selected state
- Printed ghost scale is of virtually no use to the user, and interactive rescaling is always desirable, therefore ”Fixed” should be greyed out in deselected state



NExT

Specification
Usability issues with ghost size and scale interactions (continued)

- The default height of the ghost panel should be smaller than the work area, in order that the user can immediately pick it up and move it to the zone of interest (suggest 50%)
- Setting should be exposed on style tab of ghost settings
- Setting should respond to ”Apply to similar objects” and ”Apply as default for new objects” switches